

CSE 340: Computer Architecture

[Spring 2025]

Makeup Quiz :: Set - A

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

A system uses a **direct-mapped cache** with **16 blocks**, each storing **1 Byte**. The cache starts empty. Given the following memory access sequence:

12, 20, 28, 12, 36, 44, 52, 12, 60, 68, 20, 12, 76, 84, 92

Tasks:

1. For each access, determine whether it is a **cache hit** or **miss**.
2. Show:
 - a. Byte address
 - b. Cache index
 - c. Tag
 - d. Hit or Miss
3. Calculate the **hit ratio** and **miss ratio**.

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Makeup Quiz :: Set - B

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

A system uses a **direct-mapped cache** with **16 blocks**, each storing **1 Byte**. The cache starts empty. Given the following memory access sequence:

5, 13, 21, 29, 5, 37, 13, 45, 53, 5, 61, 69, 13, 77, 85

Tasks:

4. For each access, determine whether it is a **cache hit** or **miss**.
5. Show:
 - a. byte address
 - b. Cache index
 - c. Tag
 - d. Hit or Miss
6. Calculate the **hit ratio** and **miss ratio**.